

Cognitive Systems: Neural Networks and Applied AI

Exercise 1

Basic Concepts in Machine Learning

Presented by:

Dr. Nenad Petrovic

Slides by:

Prof. Dr.-Ing. habil. Alois Knoll

Dr. Florian Walter

Topics in this Chapter

*What is machine learning...
... and how can a system learn from a teacher?*

- What is (machine) learning?
- What different types of datasets do exist?
- Which learning paradigms do result from these types?
- How does the Perceptron learning rule work?
- What is a common method for supervised learning?

What is (Machine) Learning?

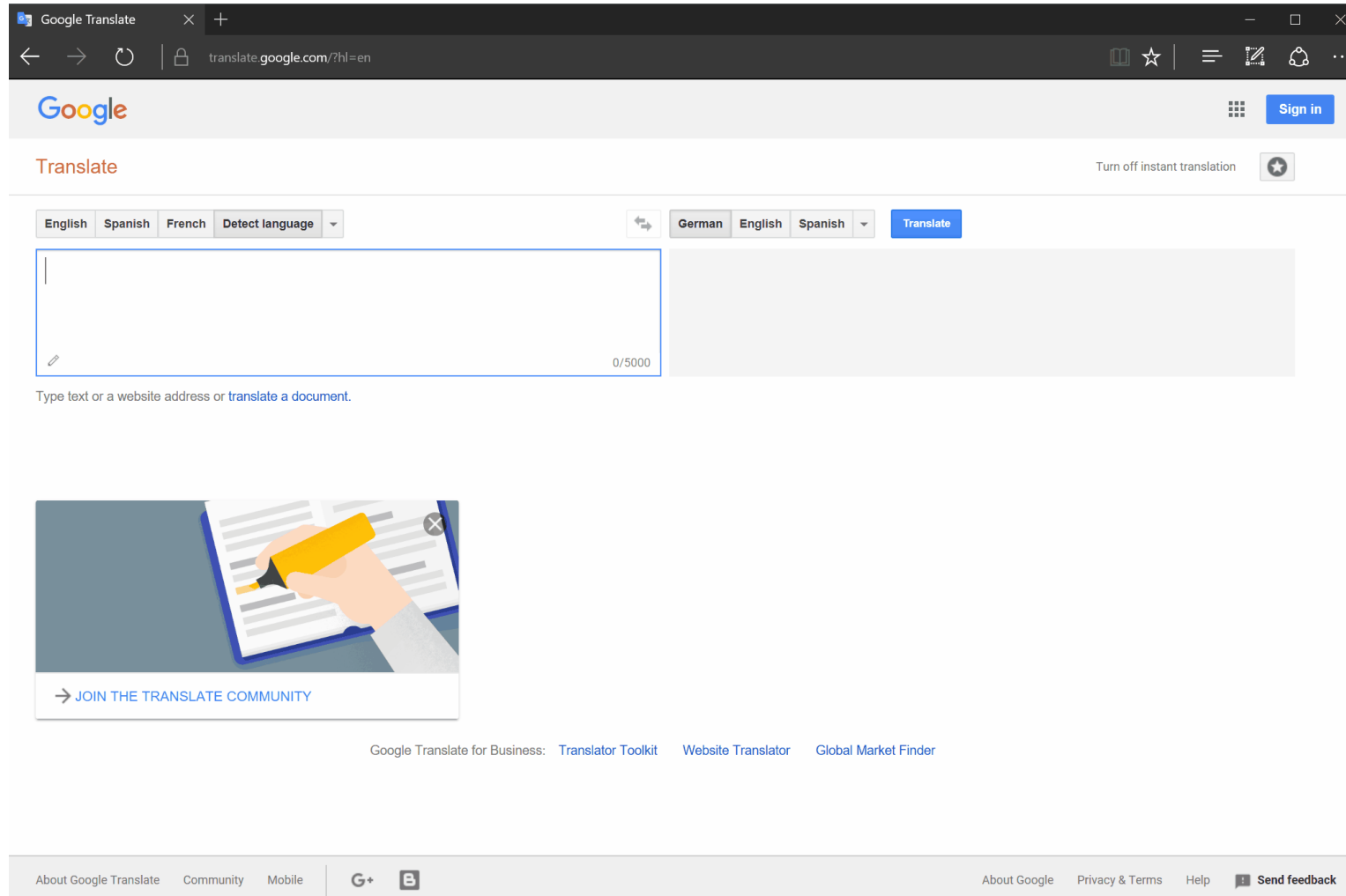
Learning – Definition

Definition E4.1: Learning

Learning is the acquisition of new information or knowledge or the process to acquire knowledge or skill by systematic study or by trial and error.

Adapted from the Encyclopedia of Neuroscience

Is this System Based on Machine Learning?



Is this System Based on Machine Learning? Yes!



Google Research Blog

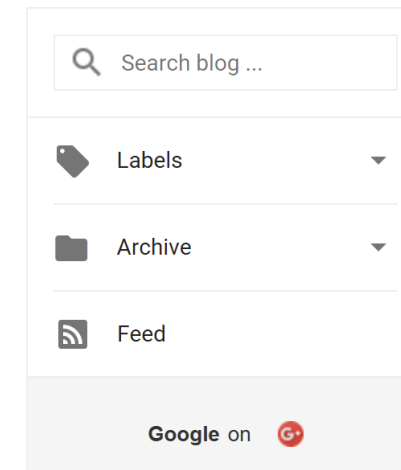
The latest news from Research at Google

Zero-Shot Translation with Google's Multilingual Neural Machine Translation System

Tuesday, November 22, 2016

Posted by Mike Schuster (Google Brain Team), Melvin Johnson (Google Translate) and Nikhil Thorat (Google Brain Team)

In the last 10 years, [Google Translate](#) has grown from supporting just a few languages to 103, translating over 140 billion words every day. To make this possible, we needed to build and maintain many different systems in order to translate between any two languages, incurring significant computational cost. With neural networks reforming many fields, we were convinced we could raise the translation quality further, but doing so would mean rethinking the technology behind Google Translate.



<https://research.googleblog.com/2016/11/zero-shot-translation-with-googles.html>

More Recently...



Is this System Based on Machine Learning?

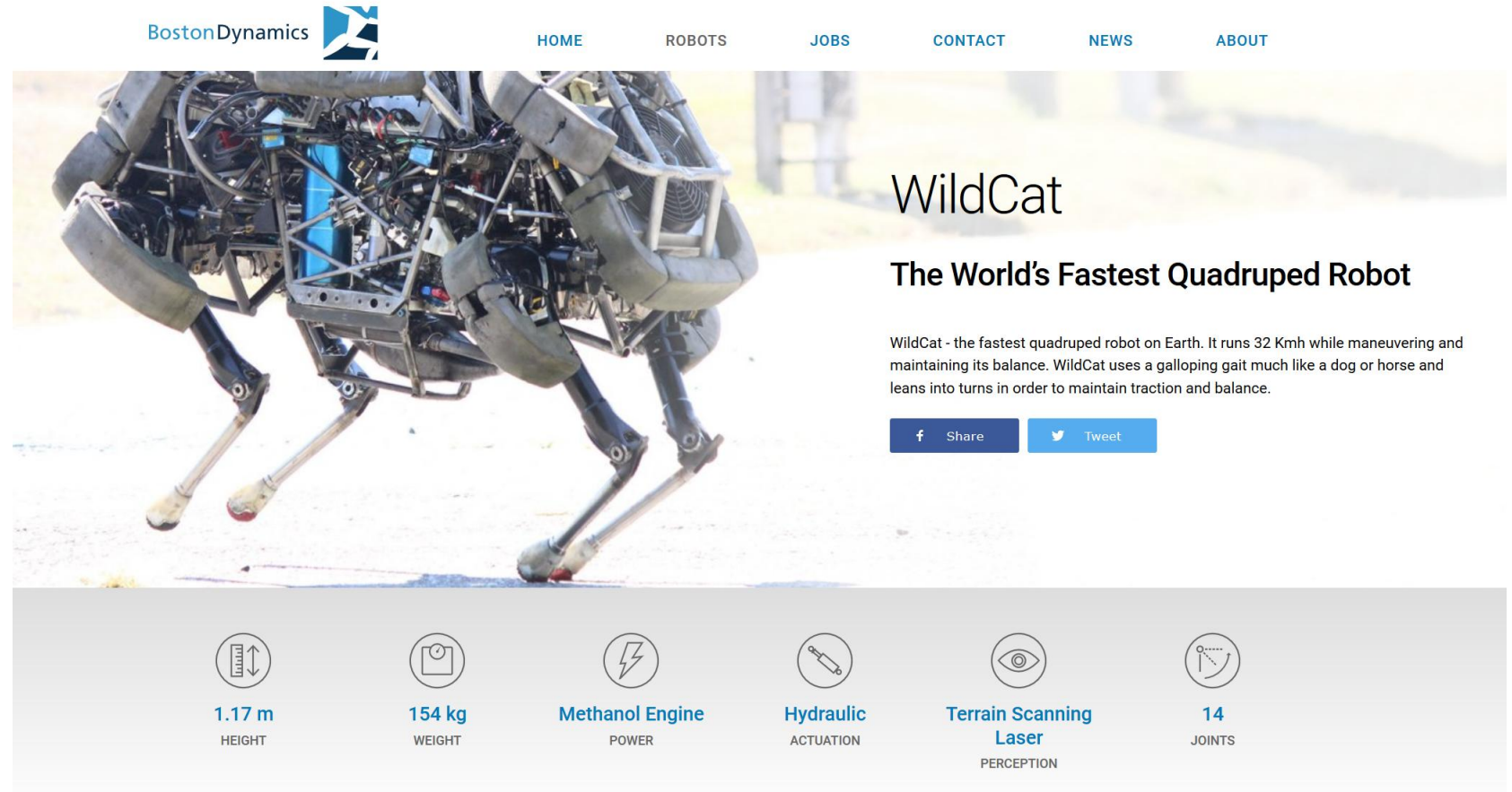


From <https://www.youtube.com/watch?v=wE3fmFTtP9g>

Is this System Based on Machine Learning? No!

“The robot uses a **variety of gaits**, including trotting, bounding, and galloping to maintain its balance while running and manoeuvring over relatively flat terrain. The on-board computer uses **dynamic control algorithms** and a variety of sensors (IMU, ground contract, proprioception, visual odometry) to control and stabilize the running motion.”

From
<https://www.bostondynamics.com/wildcat>



BostonDynamics

HOME ROBOTS JOBS CONTACT NEWS ABOUT

WildCat

The World's Fastest Quadruped Robot

WildCat - the fastest quadruped robot on Earth. It runs 32 Km/h while maneuvering and maintaining its balance. WildCat uses a galloping gait much like a dog or horse and leans into turns in order to maintain traction and balance.

f Share Tweet

					
1.17 m	154 kg	Methanol Engine	Hydraulic	Terrain Scanning Laser	14
HEIGHT	WEIGHT	POWER	ACTUATION	PERCEPTION	JOINTS

What is Machine Learning?

Machine learning is “the field of study that gives computers the ability to learn **without being explicitly programmed**” (Arthur Samuel, 1959). A machine learning system is comprised of the following four components:

- **Dataset $\mathcal{S}$** : a set of samples generated by some system or process; the samples can be single data points or pairs of input and output values
- **Model $\mathcal{M}$** : an adjustable and compact representation of a certain class of input/output relationships that is hypothesized to be capable of modeling the system or process which generates $\mathcal{S}$
- **Objective Function $\mathcal{L}$** : a function that encodes the current performance of $\mathcal{M}$ (e.g., loss or reward)
- **Algorithm $\mathcal{A}$** : the learning algorithm that adjusts $\mathcal{M}$ based on $\mathcal{S}$ and $\mathcal{L}$

Machine Learning in Artificial Cognitive Systems

Machine learning is an important prerequisite for the implementation of a broad range of cognitive functions in artificial cognitive systems:

- **Learning and Development**: modeling and implementation of biological learning mechanisms (operant conditioning, implicit learning, explicit learning, perception etc.)
- **Memory, Knowledge, and Internal Simulation**: modeling and implementation of the encoding, storage and retrieval of facts, experiences, and actions (e.g., associative memory)
- **Perception**: learning basic features to detect and categorize perceptual stimuli (e.g., unsupervised learning of visual features)
- **Autonomy**: dynamic adaption to changes in the environment (e.g., continuous online learning from a live data stream)

Programming vs. (Machine) Learning

Machine learning is an important feature of artificial cognitive systems that are situated in **complex dynamic environments**. In these environments, it is not possible to program everything in advance because...

- ... the environment is **changing continuously** (e.g., through actions of other cognitive agents or through physical processes)
- ... dynamics and objects are **too complex** to be modeled explicitly (e.g., faces, animals, plants)
- ... the **system** itself is subject to **change** (e.g., through growth, aging, injury, exercise, tool use)



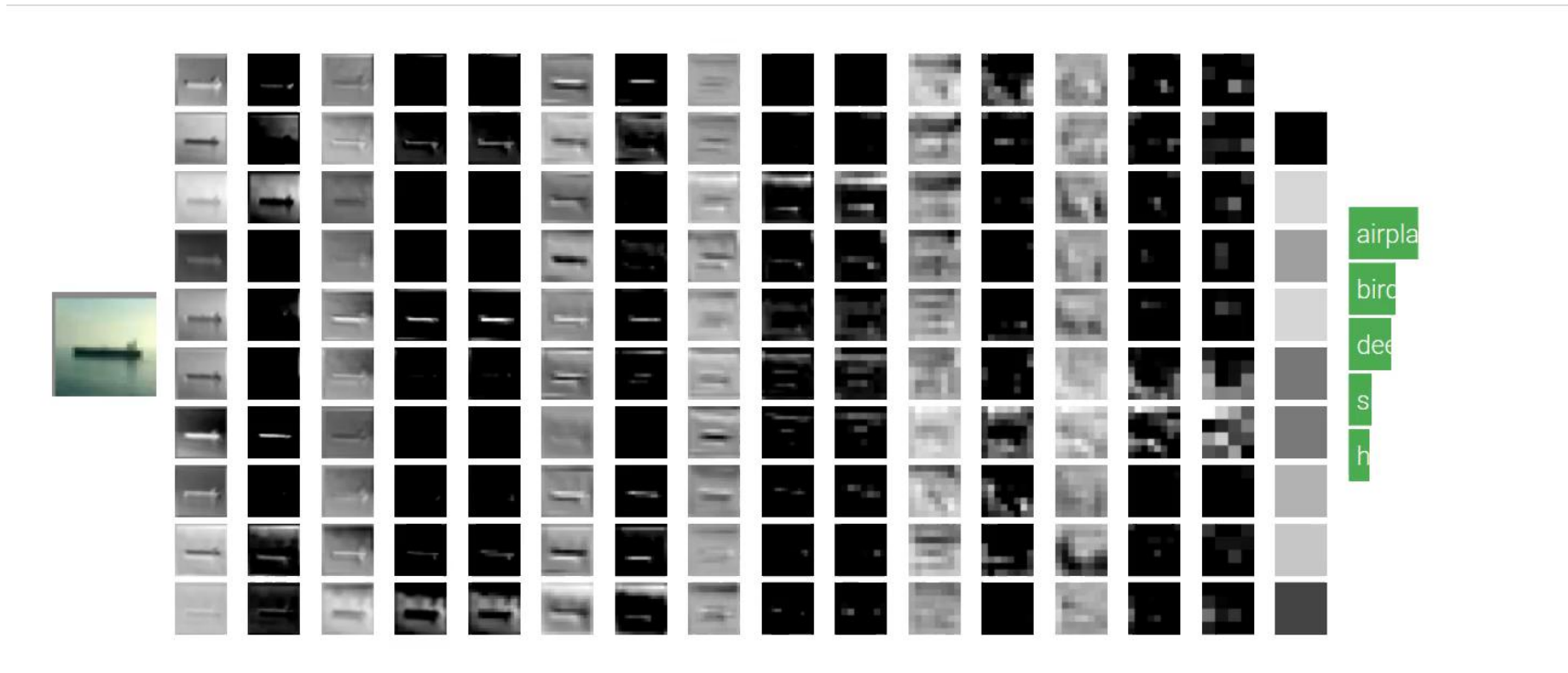
vs.



Practical Applications of Machine Learning

- Image classification
- Speech recognition
- Autonomous driving
- Recommendation systems
- Threat protection
- Control systems
- ...

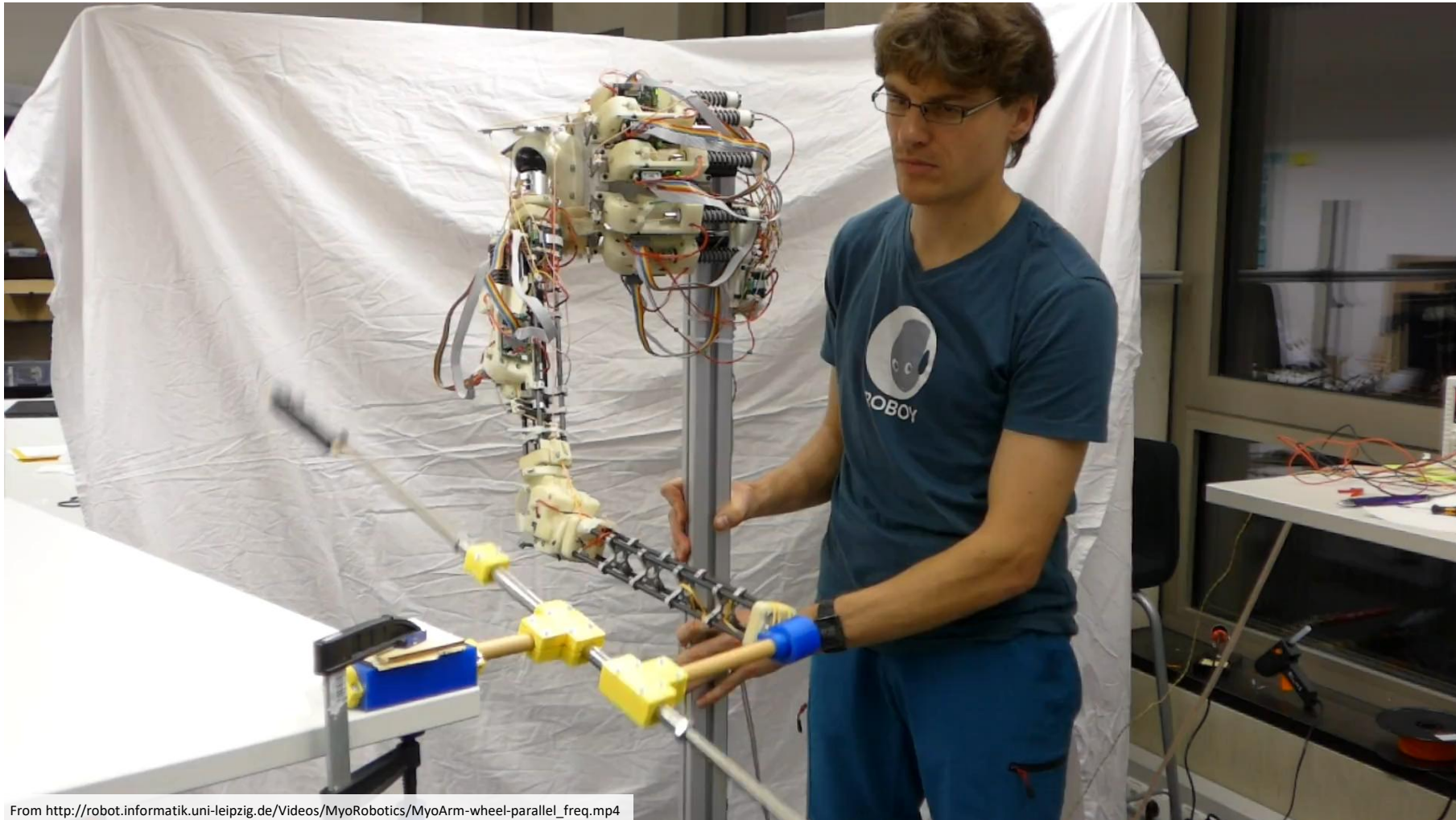
Example: Image Classification



From <http://cs231n.stanford.edu/>

Image classification of the CIFAR-10 dataset with a convolutional neural network with 17 layers and 7000 parameters.

Example: Unsupervised Learning in Robotics

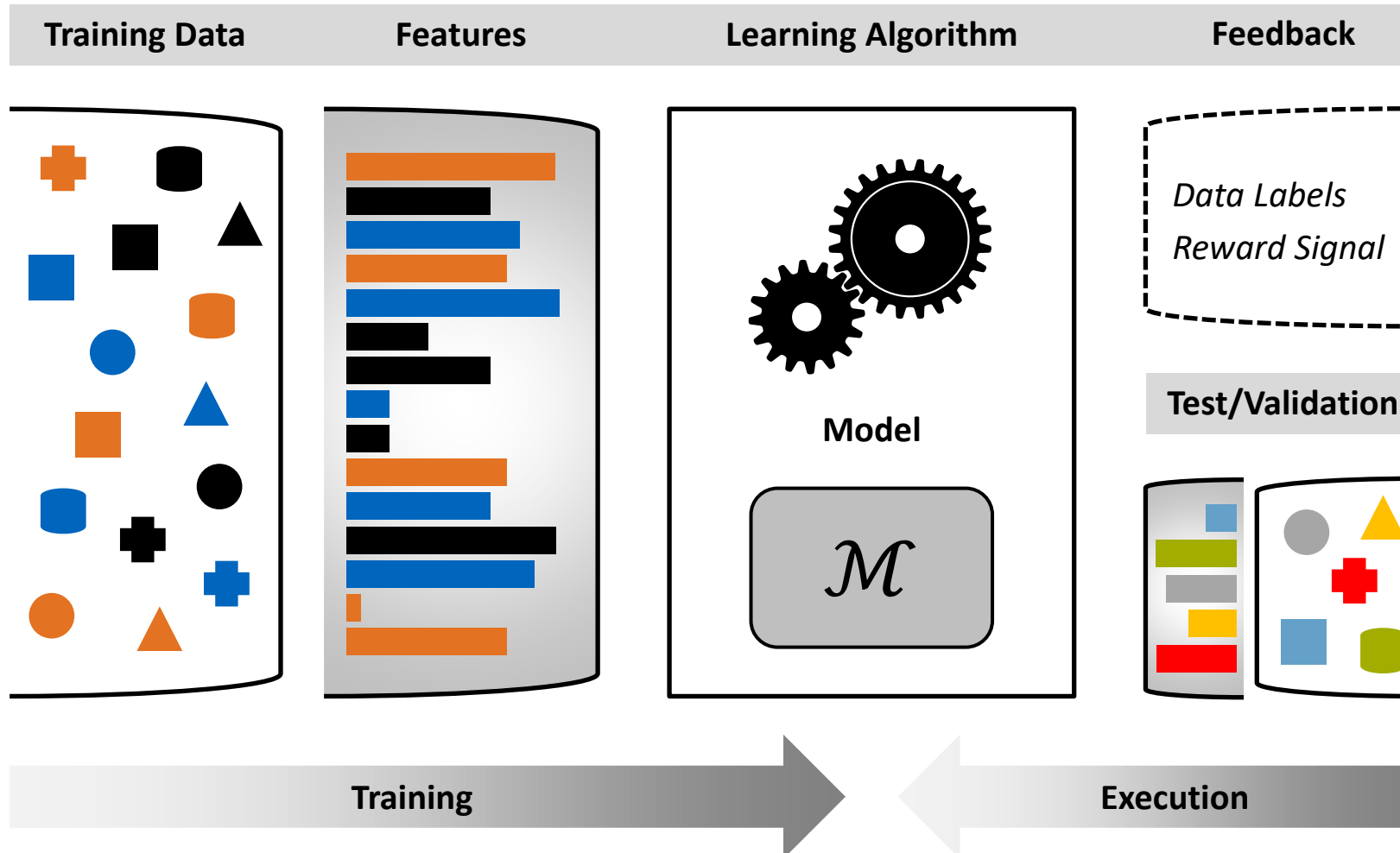


From http://robot.informatik.uni-leipzig.de/Videos/MyoRobotics/MyoArm-wheel-parallel_freq.mp4



Datasets and Learning Paradigms

Machine Learning: The General Workflow



Feature Selection and Feature Vectors

- There are **many different types** of features that can be used to describe the data at hand (shapes, colors, histograms, filters etc.)
- The selection of the right features (**feature engineering**) is critical for the performance of the machine learning model – the features must contain the information required for predictions
- All selected features are grouped into a **feature vector**:


$$\begin{pmatrix} \text{colors} \\ \text{edges} \\ \text{location} \\ \vdots \\ \text{dimensions} \end{pmatrix}$$

Definition of the Machine Learning Task

Overall Goal

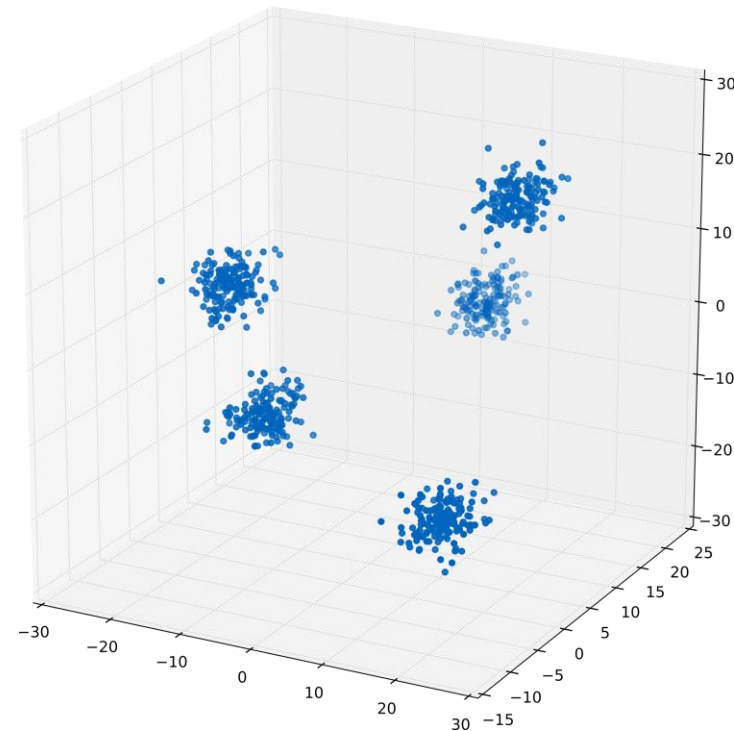
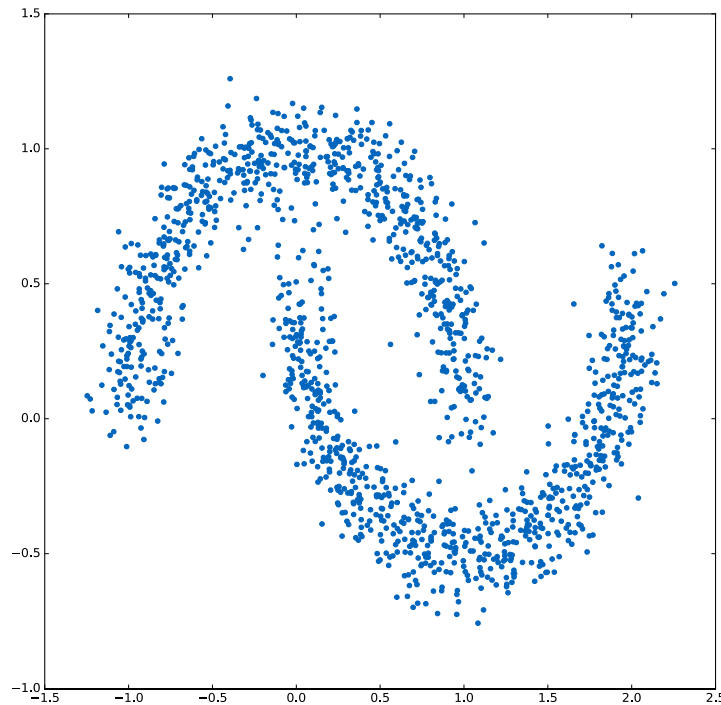
Train a model $\mathcal{M}$ in a **hypothesis space** $\mathcal{H}$ using a learning algorithm $\mathcal{A}$ so that $\mathcal{M}$ **minimizes** loss $\mathcal{L}$ for dataset $\mathcal{S}$. This type of learning is referred to as **inductive learning**.

→ The choice of $\mathcal{H}$ and $\mathcal{L}$ depends heavily on the properties of $\mathcal{S}$

Types of Datasets

- **Unlabeled data:** $\mathcal{S} = \{x_1, x_2, \dots, x_n\}$
- **Labeled data:** $\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
- **Mixed data:** $\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \cup \{x_1, x_2, \dots, x_n\}$
- **“Dynamic data”:** $\mathcal{S} = \{(x_t, s_t) \mid s_{t+1} = \mathcal{D}(x_t, s_t) \wedge 0 \leq t \leq T\}$ where $\mathcal{D}$ is a dynamical system and s_t denotes the system state

Example: Unlabeled Datasets



Most real-world data is unlabeled (e-mails, images, video recordings, measurement data) and assigning labels is usually a tedious **manual task**. However, natural datasets usually have specific **structural features** (→ **Big Data**, **Datamining**).

Example: Labeled Handwritten Digits

5	0	4	1	9	2
1	3	1	4	3	5
3	6	1	7	2	8
6	9	4	0	9	1
1	2	4	3	2	7
3	8	6	9	0	5

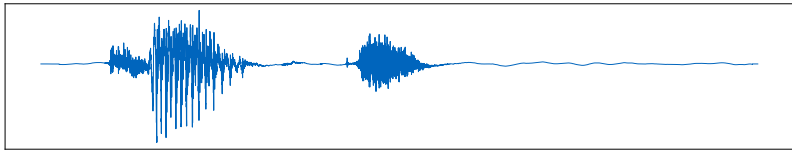
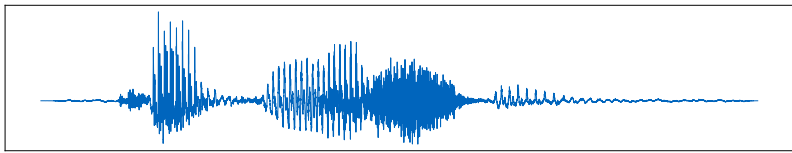
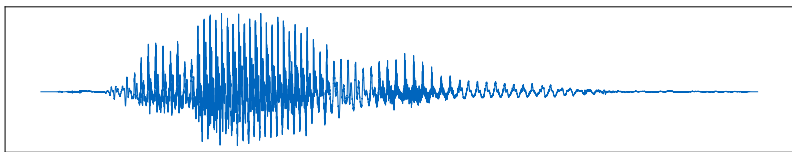
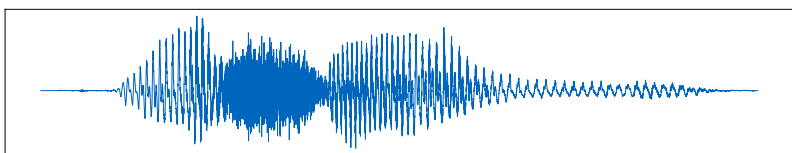
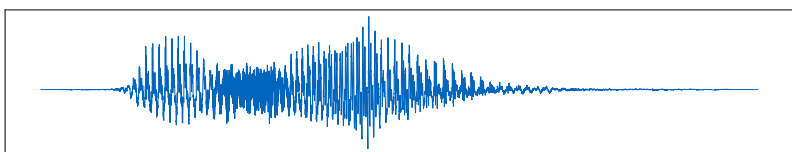
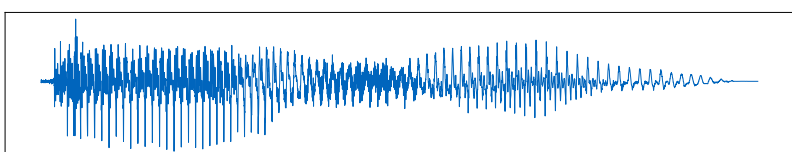
Raw Data

5	0	4	1	9	2
1	3	1	4	3	5
3	6	1	7	2	8
6	9	4	0	9	1
1	2	4	3	2	7
3	8	6	9	0	5

Labels

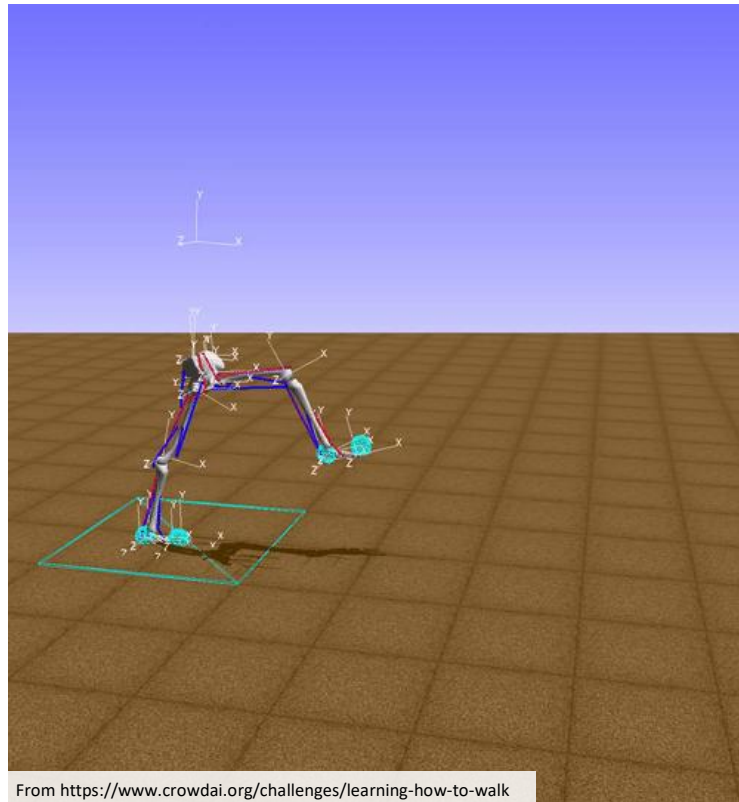
The **MNIST** dataset contains 70,000 **size-normalized**, **centered**, and **labeled** handwritten 28x28 pixel digits (<http://yann.lecun.com/exdb/mnist/>).

Example: Labeled Audio Files

		cat
		cognition
		learning
		machine
		azure
		azure

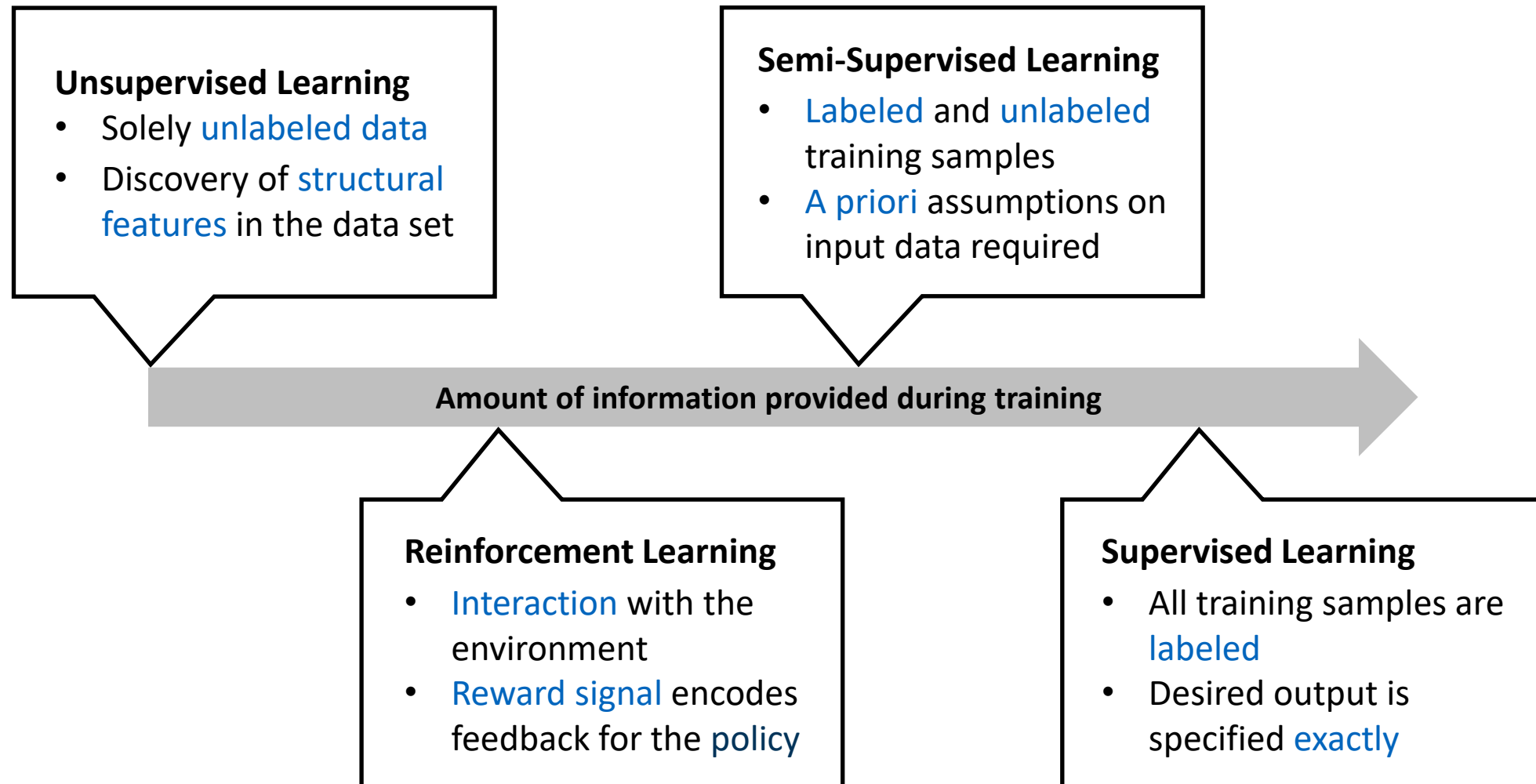
Audio from <http://www.leo.org/>

Example: Dynamic Tasks and Environments



Many real-world tasks have **complex dynamics** and an **unknown goal representation** and therefore cannot be modeled as pairs of input and desired output.

Types of Machine Learning



Datasets and Learning Paradigms

The choice of a **learning paradigm** is in most cases directly motivated by the type of data available:

Unlabeled Data	→	Unsupervised Learning
Labeled Data	→	Supervised Learning
Mixed Data	→	Semi-Supervised Learning
Dynamic Environment	→	Reinforcement Learning

- In practice, especially supervised learning requires assigning labels **manually**
- Unsupervised learning is often applied to labeled datasets as a **preprocessing** tool for initial **data analysis**